

“What distinguishes an average DevOps engineer from a proficient one is programming skills!”

Suman Debnath, Principal Machine Learning Advocate at **AWS**, attributes his first experiments with open source to his curiosity to build better tools when he was working at Toshiba. In a free-wheeling chat with *OSFY's Yashaswini Razdan*, he speaks about the need for fundamental skills such as knowledge of a basic programming language, networking and automation to build a career in DevOps.



Suman Debnath,
Principal Machine Learning
Advocate at AWS

Q. What are the career options available for a software engineer in domains where open source is important and useful?

A. There are plenty of opportunities in DevOps such as the DevOps pipeline in SaaS companies or quality engineering (QE)/quality assurance (QA) and automation. Most of the SaaS companies are built on top of some cloud vendors with a strong pipeline, which is completely automated. No one does everything manually because as a SaaS provider, scale is important. So, if you have DevOps skills, you should get the opportunity to work for companies who are on the SaaS background in their DevOps pipeline. When it comes to QA and automation for software or hardware products, most test cases are automated for any test team. As a DevOps engineer, you can build that skillset as you already know programming to automate things.

Q. Apart from DevOps, are there any other roles that software engineers can aspire for where open source plays an important role?

A. Yes, there are. Consider MLOps, a subdivision of DevOps tailored for machine learning products. If you're interested in machine learning and have a DevOps background, you can transition into MLOps roles, leveraging your understanding of automation and containerisation. MLOps is a set of practices that automate and simplify machine learning (ML) workflows and deployments. This transition offers opportunities to move from manual testing to automation and from DevOps to MLOps engineering. With

open source tools like Kubernetes and Docker, you can build pipelines for machine learning models, creating datasets, training, testing, and deploying models efficiently.

Moreover, open source is fundamental across various domains within computer science. For instance, proficiency in containers or programming languages like Python opens numerous opportunities and allows you to adapt to different roles within the tech industry.

What distinguishes an average DevOps engineer from a proficient one is programming skills. While many DevOps professionals excel in automation, aspiring for roles in software development requires